Porphyria Cutanea Tarda (PCT)

Contents

This information on PCT is based on best available evidence and the consensus of a subgroup of porphyria specialists in the International Porphyria Network (Ipnet).

1. Introduction

Porphyria cutanea tarda (PCT) is the commonest porphyria worldwide, affecting between 1:5,000 to 1:70,000 of the population, depending on geographical location. PCT presents predominantly in adulthood ('tarda' means 'late').

PCT occurs due to reduced activity of hepatic uroporphyrinogen decarboxylase (UROD), the fifth enzyme in the haem biosynthesis pathway. This is a consequence of an iron catalyzed reaction which produces porphomethene, a UROD inhibitor. The reduced activity of UROD leads to increased production of porphyrin intermediates in the liver, which are released into the systemic circulation and accumulate in the skin, causing cutaneous fragility and blistering to photoexposed sites. Clinically significant PCT symptoms occur when hepatic UROD activity is <20%.

2. Genetics and epidemiology

PCT affects both men and women, with a slight male preponderance in Northern Europe and the United States but affects predominantly males in southern Europe due to increased exposure to risk factors. Clinical presentation is usually over the age of 40, peaking in the 5th and 6th decades with earlier presentation suggesting familial PCT (see below).

PCT is classified into 3 subtypes:

- **Sporadic or type 1 PCT** is not inherited and in Europe is the most common form (about 80% of cases). It typically presents in middle age as a consequence of the predisposing risk factors described below.
- **Familial or type II PCT** is an autosomal dominant condition with low clinical penetrance. Inheritance of a genetic mutation inactivates one UROD allele, reducing UROD activity in all tissues to about 50% of normal. As with other porphyrias there is wide allelic heterogeneity. Clinical penetrance within families is approximately 10-20% as a further reduction in hepatic UROD activity to below 20% is required for clinical presentation. The additional decrease in UROD activity in familial PCT is also a consequence of the triggering factors described below.

- Hepatoerythropoietic porphyria (HEP) is a very rare homozygous form of familial PCT, with only around 40 cases reported worldwide. It is characterized by severe photosensitivity, usually presenting in early childhood. Most patients are homozygous or compound heterozygous for missense genetic mutations. The majority of porphyrins are produced in the liver, and most patients are haematologically normal.

3. Pathogenesis and precipitating factors

Both familial and sporadic PCT result from conditions that induce progressive inhibition of UROD in the liver, resulting in the accumulation of uroporphyrinogen, and its partially decarboxylated intermediates. The porphyrinogens undergo oxidation to their equivalent porphyrins, which act as chromophores in the skin and absorb light at wavelengths of 405 nm (the Soret band), which is able to penetrate into the dermis and basement membrane. Porphyrins circulating through dermal blood vessels are photosensitised on exposure to longer wavelength UVA and visible light and emit free radicals that result in the formation of reactive oxygen species (ROS) which damage the dermo-epidermal junction, leading to the formation of subepidermal bullae.

In most cases PCT is associated with mild abnormal liver function, which may be evident on routine blood testing, and is caused by a number of identifiable precipitating factors which can result in higher oxidative stress to hepatocytes and/or increased iron absorption caused by downregulation of hepcidin. Most patients will have at least one, but occasionally no predisposing factors are identified.

Precipitating factors:

- **Hepatic iron accumulation:** Almost all PCT patients have an increase in hepatic iron, which is key to inactivation of the UROD enzyme. Some of these patients will have inherited genetic haemochromatosis (resulting from autosomal recessive HFE gene mutations, C282Y and H63D).
- **Excessive alcohol consumption**.

- **Hepatitis C** is an important predisposing factor for PCT in many European countries.
- **HIV infection** is associated with PCT and patients are often co-infected with hepatitis C.
- **Oestrogen therapy**. For example as oral contraception or hormone replacement therapy (HRT).
- **Renal dialysis**
- **Cigarette smoking.** Polycyclic aromatic hydrocarbons (PAH) are among the toxins produced by cigarette smoke, triggering production of CYP1A2 which stimulates UROD inhibition.
- **Exposure to some industrial chemicals**
- **Rare associations:** Diabetes, systemic lupus erythematosus, chronic renal disease and haematological malignancy.

4. Clinical features

Photoexposed sites are affected, most frequently the dorsal hands, forearms, face and scalp. As a result of skin fragility even minor injury or everyday tasks involving the exposed areas may cause skin tears, erosions and large non-inflammatory blisters. The bullae heal slowly with scarring and milia (small harmless keratin inclusion cysts). Both hyper and hypopigmentation can occur as can hypertrichosis, particularly affecting the face.

Urine may become strikingly dark due to excess uroporphyrin excretion.

Less common clinical features include patchy or diffuse sclerodermatous changes, which may affect non-exposed areas of skin, scarring alopecia and onycholysis.
Acute attacks (abdominal pain and neurological symptoms, often provoked by drugs or hormones), that occur in the acute porphyrias are not a feature of PCT.

5. Laboratory Diagnosis

Skin biopsies are not required.

Porphyrin testing:
PCT is diagnosed by identifying the pattern of excess porphyrins in samples of blood, urine and faeces. Samples should be analysed by an accredited laboratory approved for quantifying porphyrins and identifying patterns of individual porphyrins.

Key diagnostic features include:

- *Plasma porphyrin* – a strongly positive plasma porphyrin fluorescence emission scan with peak emission at 615-620 nm.

- *Urine porphyrin* – greatly increased urine porphyrin excretion (10- 20 X's) with predominantly uroporphyrin and heptacarboxylic porphyrin evident on chromatography.
- *Faecal porphyrin* – usually increased 2-5 X's with predominantly heptacarboxylic and other partially decarboxylated intermediates. The presence of isocoproporphyrins are pathognomonic for PCT

Additional Investigations:

1. Liver function tests (LFT's)
2. Full blood count (FBC)
3. Hepatitis C (HCV) serology
4. Human Immunodeficiency Virus (HIV) serology
5. Iron studies (ferritin, transferrin saturation)
6. Haemochromatosis (HFE genotype)

Other investigations as required by clinical or laboratory findings – e.g. liver imaging, fibroscan.

6. Management of PCT

The aim of treatment is to produce biochemical and clinical remission by reducing porphyrin excretion to within normal range. Known predisposing factors must be controlled or treated which may require a multidisciplinary approach (e.g. hepatology, infectious diseases). With appropriate treatment, the cutaneous features resolve slowly often with scarring. Occasionally skin symptoms may worsen and the urine can darken when treatment is started. Relapse >2 years after successful treatment can occur and patients should therefore be followed-up or monitored after remission.

Specific primary treatment
The aim of PCT treatment is to reverse inactivation of the UROD enzyme and remove (chloroquines) or decrease production (venesection) of excess porphyrins.

Two specific treatments are available, and the modality is dictated by the predisposing factors and patient suitability.

- Venesection (phlebotomy) aims to reduce hepatic iron stores without inducing anaemia. This is generally achieved by removing a unit of blood (approximately 450 mls) every 2-4 weeks but can be more frequent (weekly) if tolerated. Treatment should continue until iron depletion is achieved, which can be monitored by transferrin saturation (<16%) and ferritin (<25 ug/L). Clinical remission usually takes 6-9 months and complete biochemical resolution approximately 12 months. Patients with HFE gene mutations may require longer-term intermittent phlebotomy.

- Low-dose chloroquine (125 mg twice weekly) or hydroxychloroquine (100 mg twice weekly). It is important that patients are advised not to take these more frequently, as larger doses can cause an acute hepatotoxic reaction. Despite low dose chloroquines having a decreased risk of retinopathy, standard ophthalmological guidelines for monitoring should still be followed. Chloroquine and hydroxychloroquine release porphyrins from hepatocytes, increasing urinary excretion, which may result in some darkening of the urine. Treatment should be continued until biochemical remission has been established for a few months. Clinical resolution usually occurs within 6 months, with biochemical resolution taking up to 12 months. This treatment is contraindicated in advanced liver and renal disease.

- The above treatments can be combined.

Other options in complex or refractory cases include iron chelation e.g. desferrioxamine. In renal failure, where patients may be unable to tolerate either treatment, erythropoietin without iron supplementation may be effective, and can be supplemented with small volume venesection.

General measures

- **Photoprotection** is extremely important, particularly when cutaneous symptoms are present. Patients should be advised to wear protective clothing, including wide brimmed hats, long sleeves and trousers to limit their exposure to sunlight. As skin damage is caused by the visible light, UV sunscreens are not effective. Physical sunscreens based on titanium dioxide or zinc oxide may be helpful.
- Protection of fragile skin from ongoing damage is also important and patients should be advised to wear protective clothing such as gloves when undertaking work related or household tasks. Wounds should be kept clean, and any skin infection treated promptly. Blister care, including piercing of new blisters with a sterile needle and leaving the roof in place, should be explained to the patient. Deroofing of blisters must be avoided as this increases the risk of secondary cutaneous infection.
- Serum vitamin D levels should be monitored due to photoprotection limiting vitamin D production in the skin. Dietary sources are highly unlikely to provide sufficient levels of vitamin D and a daily supplementation of 800-1000 IU is recommended.

Other measures

- **Alcohol**. All alcoholic drinks should be avoided. Complete abstention may result in clinical remission
- **Cigarette smoking** should be stopped.
- **Oestrogen therapy**. Withdrawal of oestrogen alone may result in remission, particularly if oestrogen use has been short-term. Women on oestrogen treatments will be asked to stop

taking them while the PCT is being treated. Once the PCT has been treated it may be possible to restart hormone treatment, preferably via a transdermal route (e.g. HRT).

- **Iron supplements** should not be taken, unless there is clear evidence of iron deficiency.
- **Treatment for hepatitis C.** PCT in patients with hepatitis C may resolve on treatment with direct acting antiviral (DAA) drugs. Specific treatments may therefore not be required.

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An international network of experts working together to improve healthcare services for porphyria patients

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